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Method for stabilizing thermoplastic recyclates, stabilized plastic compositions, and molding compounds and molded parts produced therefrom

Abstract

The invention relates to a method for stabilizing thermoplastic recyclates or previously damaged plastic materials against oxidative, thermal and/or actinic degradation. According to the claimed method, at least one substituted sugar, selected from the group consisting of at least one ester and/or ether of an alditol or cyclitol, at least one ester of a saccharic acid, at least one amide and/or substituted amine of an amine sugar, and mixtures and combinations thereof, as the stabilizing component is introduced in a thermoplastic recyclate and optionally additionally at least one primary antioxidant and/or at least one secondary antioxidant is introduced in a thermoplastic recyclate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is the U.S. national phase of International Application No. PCT/EP2020/058188, filed on Mar. 24, 2020, which claims the benefit of German Patent Application No. 10 2019 204 160.4, filed Mar. 26, 2019, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

BACKGROUND OF THE INVENTION

(2) The present invention relates to a method for the stabilization of thermoplastic plastic recyclates or pre-damaged plastics against oxidative, thermal, and/or actinic degradation. In the method in accordance with the invention, at least one substituted sugar, selected from the group consisting of at least one ester and/or ether of an alditol having at least four carbon atoms or a cyclitol, at least one ester of a saccharic acid, at least one amide and/or substituted amine of an amino sugar, and mixtures and combinations thereof, is/are introduced as a stabilizing component into a

thermoplastic plastic recyclate and, optionally additionally hereto, at least one primary antioxidant and/or at least one secondary antioxidant. Plastic recyclates can be stabilized against oxidative, thermal, and/or actinic degradation with a high efficiency and in a very environmentally friendly and inexpensive manner by the method in accordance with the invention. The present invention additionally also relates to corresponding recyclate-based plastic compositions and to molding compounds and molded parts produced therefrom. The present invention furthermore also relates to stabilizer compositions and to their use for the stabilization of thermoplastic plastic recyclates against oxidative, thermal, and/or actinic degradation. Plastic recyclates are a growing market and an important element in recycling management of natural resources, with recyclates then ideally being intended to replace new plastics having an identical or at least comparable property profile. However, with recyclates from production or from waste plastics, irreversible changes in the polymer chain occur, often over many years, due to mechanochemical, chemical, or light induced processes during the first processing steps (e.g. compounding, injection molding) and during use, (see e.g. R. Pfaendner et al., *Angew. Makromol. Chemie* 1995, 232, 193-227; Pospisil et al., *Pol. Degr. Stab.* 1995, 48, 351-358). New chemical groups are produced in the polymer chain and/or the composition of the polymer is changed by radical reactions in the presence of oxygen. The concentration of these newly formed groups increases with the length of use and the area of use (UV light, high temperatures, contact media). The structural inhomogeneities of a recyclate then influence the plastic properties such as the mechanical and physical properties and the processing behavior.

(3) It furthermore applies that recyclates or pre-damaged plastics are more sensitive to oxidation than new material since degradation products act as initiator sites for a further oxidation or as prodegradants (A. S. Maxwell, *Pol. Eng. Sci.* 2008, 381-385; I. H. Craig, J. R. White, *J. Mater. Sci.* 2006, 41, 993-1006, L. Luzuriaga, J. Kovarova, I. Fortelny, *Pol. Degr. Stab.* 2006, 91, 1226-1232). Post-stabilization with selected stabilizers is an important method to achieve a quality improvement of plastic recyclates. The stabilizers used protect the recyclate from further oxidative (or photo-oxidative) damage. The optimized stabilizer composition for a recyclate, however, differs due to the described structural deviations from the stabilization composition of the new product (see e.g. R. Pfaendner, *Kunststoffe International* 12/2015, 41-44).

(4) Due to the differences now acknowledged between new plastic products and plastic recyclates, special recyclate stabilizer compositions have become known and are also available as commercial products (e.g. the Recyclostab series of BYK (Chemie GmbH, Wesel). Known technical stabilization solutions that take account of the demands of recyclates are, for example: Stabilization composition consisting of a phenolic antioxidant, a phosphite, and a fatty acid salt (EP 0662101). Stabilization composition consisting of a phenolic antioxidant, a phosphite, and a metal oxide such as calcium oxide (U.S. Pat. Nos. 6,525,158, and 6,251,972). Stabilization composition consisting of a phenolic antioxidant and a polyfunctional epoxide (EP 0702704). Stabilization by a macrocyclic piperidine (U.S. Pat. No. 5,789,470). Stabilization composition consisting of a secondary aromatic amine and a polyfunctional epoxide (WO 97/30112). Discoloration of damaged polyolefins by a hydroxylamine (EP 470048). Mixture of antioxidants, mold lubricants, anti-blocking agents, UV stabilizers, and antistatic agents for recyclate films (DD 288161). Stabilization composition for mixed plastics consisting of a phenolic antioxidant and a phosphite/phosphonite (EP 0506614). Stabilization of polycarbonate blends comprising post-consumer polystyrene recyclates by acids (WO 2014/141178). Stabilization of halogenated recyclates by 6-aminouracils (EP 1 426 406)

(5) Despite the known solutions, there is still a need for particularly high performance or effective, environmentally friendly, and inexpensive options for the stabilization of plastic recyclates or stabilizer compositions usable therefor.

(6) Recyclates have here already run through a life cycle and had already been compounded to form molded plastic parts and as such exposed to mechanical, chemical, actinic, and/or oxidative loads. Specific defects hereby result in the polymer chain that effect further, disproportionate

damage to the underlying plastic material on a repeat processing of the recyclates. The damage points present in the polymer here serve as initiator points for an accelerated and expanded degradation of the polymer. A lowering of the molecular weight, a broadening of the molecular weight distribution, a reduction in the mechanical properties, and an increase in the light sensitivity are in particular typical phenomena that occur with recycled plastics, in particular on their repeat processing. The property profile that has worsened overall is thus more disadvantageous for a repeat processing.

SUMMARY OF THE INVENTION

(7) Starting from this, it was thus the object of the present invention to provide a method for the stabilization of thermoplastic plastic recyclates by which the plastic recyclates can be stabilized against oxidative, thermal, and/or actinic degradation in a very effective, environmentally friendly and inexpensive manner.

(8) This object is achieved by a method for the stabilization of thermoplastic plastic recyclates described herein, by the features of the plastic composition described herein, by the features of the molded part manufactured from the plastic composition described herein, by the features of the stabilizer composition described herein, by the use of a substituted sugar and the use of the stabilizer compositions also described herein, as well as the advantageous further developments also described herein.

Description

DETAILED DESCRIPTION OF THE INVENTION

(1) In accordance with the invention, a method is thus provided for the stabilization of thermoplastic plastic recyclates against oxidative, thermal, and/or actinic degradation in which at least one substituted sugar, selected from the group consisting of (a1) at least one ester and/or ether of an alditol having at least four carbon atoms or a cyclitol; (a2) at least one ester of a saccharic acid; and/or (a3) at least one amide and/or substituted amine of an amino sugar, and mixtures and combinations thereof (also called “component (A)” in the following) is/are introduced into the thermoplastic plastic recyclate.

(2) The substituted sugars can respectively be chemically pure esters, ethers, and amides. It is equally possible that the respective substituted sugars have a mixed substitution pattern. It is, for example, possible that a substituted sugar comprises both an ester functionalization and an amide functionalization.

(3) The substituent(s) of the substituted sugars can be aliphatic, aromatic, or cycloaliphatic hydrocarbon groups, optionally interrupted by oxygen atoms, sulfur atoms, or nitrogen atoms; groups having at least four, particularly preferably having at least six, CH₂ groups are preferred.

(4) The preparation of the substituted sugars can take place in accordance with known methods that are e.g. described in the following citations: E. Reinefeld, G. Klaunberg, *Tenside* 1968, 5 (9-10), 266-270 DE 2423278, B. V. Chemie Combinatie Amsterdam C.C.A. A. F. Artamanov et al. *Chemistry of Natural Compounds*, 2001, 37, 124-125 A preferred preparation method is here regioselective via enzymatic processes: Y. Watanabe et al., *Tetrahedron* 2013, 69, 4657-4664 W. Wei et al. *Green Chemistry* 2015, 17, 3475-3489

(5) Since the thermoplastics forming the basis of the plastic recyclates had as a rule already had primary and/or secondary antioxidants on their manufacture, it is therefore not absolutely necessary in the reprocessing of consumed or already used plastics, that is recyclates, to again add primary and/or secondary antioxidants in the reprocessing of these recyclates. It has surprisingly been found that it is sufficient to incorporate at least one substituted sugar such as defined above into the plastic recyclates to achieve a stabilizing effect that is particularly high in a synergetic manner in

the presence of primary and/or secondary oxidants.

(6) In accordance with the present invention, a plastic recyclate is understood as a recycled plastic in accordance with the definition of the term as per the standard DIN EN 15347:2007. Furthermore, relevant international standards exist for many kinds of plastic recyclates. DIN EN 15353:2007 is, for example, relevant to PET plastic recyclates. PS recyclates are described in more detail in DIN EN 15342:2008. PE recyclates are treated in DIN EN 15344:2008. PP recyclates are characterized in DIN EN 15345:2008. For the purposes of the corresponding specific plastic-recyclates, the present invention incorporates the definitions of these international standards.

(7) Unlike new plastics, plastic recyclates usually have pre-damage, i.e. new chemical groups are produced on the polymer chain by oxidative or (photo) oxidative processes. With polyolefins, these are e.g. carbonyl groups that are not present or are only present to a highly subordinate degree with new products. The concentration of the carbonyl groups is therefore simultaneously a measure for the pre-damage of the polymer. The concentration of carbonyl groups can be determined in accordance with known analytical methods such as infrared spectroscopy, such as is described in E. Richaud et al. *Pol. Degr. Stab.* 2009, 94, 410-420. In this case, the absorption of the carbonyl vibration in the region of 1720 cm⁻¹ is measured.

(8) A preferred embodiment of the method in accordance with the invention provides that the at least one substituted sugar comprises three, preferably at least four, free OH groups.

(9) It is equally advantageous if the at least one substituted sugar is mono- or disubstituted.

(10) Monosubstituted compounds are very particularly preferred.

(11) In accordance with a preferred embodiment, the alditol from which the substituted sugars are derived has the molecular formula

HOCH.sub.2[CH(OH)].sub.nCH.sub.2OH,

R.sub.1—OCH.sub.2[CH(OH)].sub.nCH.sub.2OH, or

HOCH.sub.2[CH(OH)].sub.n[CH(OR.sub.1)]CH.sub.2OH

where n=2-6, preferably n=3-5, where R.sub.1 is an optionally substituted sugar residue.

(12) The at least one alditol is preferably derived from alditols selected from the group consisting of threitol, erythritol, galactitol, mannitol, ribitol, sorbitol, xylitol, arabitol, ismaltol, lactitol, maltitol, maltotritol, and hydrated oligosaccharides and polysaccharides with polyol end groups and mixtures thereof. The at least one preferred alditol is particularly preferably selected from the group consisting of erythritol, mannitol, isomaltol, maltitol, and mixtures thereof. The at least one alditol is very particularly preferably mannitol or the cyclitol is myo-inositol.

(13) Examples for heptitols and octitols as starting products for the substituted derivatives are: meso-glycero-allo-heptitol, D-glycero-D-altro-heptitol, D-glycero-D-manno-heptitol, meso-glycero-gulo-heptitol, D-glycero-D-galacto-heptitol (perseitol), D-glycero-D-gluco-heptitol, L-glycero-D-gluco heptitol, D-erythro-L-galacto-octitol, D-threo-L-galacto-octitol. Alternatively or additionally to the above-named alditols, cyclitols, i.e. ring-shaped polyols, can also be used as starting materials for the preparation of the substituted sugars in the method in accordance with the invention (or for the purposes of the plastic composition, the molding compound, or the molded part or the stabilizer composition in accordance with the invention).

(14) The at least one cyclitol for the starting material for the preparation of the substituted sugars is further advantageously selected from the group consisting of inositol, quercitol, viscumitol, bornesitol, conduritol, ononitol, pinitol, prinpolitol, cerceritol, querbrachitol, quinic acid, shikimic acid, and valienol, with the at least one cyclitol preferably being myo-inositol.

(15) The at least one ester of the saccharic acid is further advantageously an aldonic acid or an aldonic acid. Starting materials for the preparation of the substituted saccharic acids are, for example, the aldonic acids allonic acids altronic acid, gluconic acid, mannonic acid, gulonic acid, idonic acid, galactonic acid, talonic acid, ribonic acid, arabonic acid, xylonic acid, lyxonic acid, erythronic acid, or threonic acid. Examples for preferred aldonic acids are mucic acid or glucaric acid.

(16) The at least one substituted amine or amide is further advantageously an amino sugar. Starting materials for the preparation of the substituted amino sugars are, for example 1-amino-1-deoxy-alditols such as 1-amino-1-deoxy-galactinol, 1-amino-1-deoxy-mannitol, 1-amino-1-deoxy-sorbitol, 1-amino-1-deoxy-maltitol. Examples for disubstituted amino sugars are e.g. derived from alpha-omega-diamino-alpha-omega-dideoxyalditols; 1,6-diamino-1, 6-dideoxysorbitol is an example.

(17) Preferred substituents are linear or branched hydrocarbon chains having at least six CH groups such as hexyl, heptyl, octyl, nonyl, decyl, isodecyl, lauryl, stearyl, myristyl, oleyl, cetyl, ceryl, myricyl, palmitoleyl, arachidyl, behenyl, eruycl, lignoceryl, montanyl, linoleyl, isotridecyl.

(18) The at least one substituted sugar is in particular selected from the group consisting of the following compounds:

(19) ##STR00001## ##STR00002##

where n is the same or different on every occurrence and is a whole number between 1 and 100, preferably 2-50, very particularly preferably 3-20.

(20) 0.01 to 5.0 parts by weight, preferably 0.02 to 1.0 parts by weight, of the at least one substituted sugar (component (A)) are in particular introduced relative to the 95.0 to 99.99 parts by weight, preferably 99.0 to 99.98 parts by weight, of the thermoplastic plastic recyclate.

(21) A preferred embodiment of the method provides that in addition to the at least one substituted sugar, at least one primary antioxidant and/or at least one secondary antioxidant (B)) is/are introduced into the thermoplastic plastic recyclate. Components (A) and (B) can here be introduced into the plastic recyclate individually or separately from one another, or components (A) and (B) can be introduced into the plastic recyclate together in the form of a composition comprising components (A) and (B) or in the form of a composition consisting of components (A) and (B). At least one primary antioxidant (component (B1)), or at least one secondary antioxidant (component (B2)), or at least one primary antioxidant and at least one secondary antioxidant is used as component (B).

(22) In the event that at least one primary antioxidant and at least one secondary antioxidant are used, they can also be introduced into the plastic recyclate individually or separately from one another or together in the form of a mixture.

(23) Component (A) used in accordance with the invention and optionally component (B) used in addition to component (A) acts or act together as a stabilizer for the thermoplastic plastic recyclate, with the oxidative, thermal, and/or actinic degradation of the thermoplastic plastic recyclate being inhibited or prevented. In other words, the plastic is stabilized against oxidative, thermal, and/or actinic degradation by the introduction of component (A) and optionally additionally of component (B) into the thermoplastic plastic recyclate.

(24) The present invention is here particularly characterized in that the at least one substituted sugar is used alone or together with primary and/or secondary antioxidants for the stabilization of thermoplastic plastic recyclates.

(25) It was surprisingly found in accordance with the invention that a very effective stabilization of thermoplastic plastic recyclates against oxidative, thermal, and/or actinic degradation can be achieved by the introduction of at least one substituted sugar alone or together with at least one primary and/or at least one secondary antioxidant. Substituted sugars are additionally very environmentally friendly and inexpensive compounds. Thermoplastic plastic recyclates can thus be stabilized against oxidative, thermal, and/or actinic degradation by the method in accordance with the invention in a very effective, environmentally friendly, and inexpensive manner.

(26) The inventors surmise here that the substituted sugars incorporated into the plastic recyclate can here react with the already present damage sites (as a rule carbonyl groups) to form hemiacetals or full acetals when processing the new compounding. The damage sites in the plastic are thus effectively masked and are no longer available as initiator sites for a further degradation. Damage points in plastic recyclates can thus be eliminated by the method in accordance with the invention

or by the additive compositions in accordance with the invention. It is further presumed that the substituted sugars in accordance with the invention can act as hydroperoxide decomposers and/or as metal deactivators.

(27) For this reason, the addition of a substituted sugar to a plastic recycle is also alone an effective stabilizer since it can frequently be assumed that primary and secondary antioxidants present from the first application in the recycle had not been completely consumed during the use.

(28) The thermoplastic plastic recycle used in the method in accordance with the invention is preferably a polyolefin recycle, e.g. a polypropylene recycle or a polyethylene recycle. The method in accordance with the invention is furthermore particularly suitable for the stabilization of pre-damaged recycles, in particular polyolefin recycles, whose carbonyl group content, determined by absorption in the infrared spectrum, amounts to at least 0.01 mol/kg, is preferably greater than 0.02 mol/kg, and is particularly preferably greater than 0.05 mol/kg.

(29) In a preferred variant of the method in accordance with the invention, the at least one primary antioxidant is selected from the group consisting of phenolic antioxidants, amines, lactones, and mixtures thereof.

(30) There can, for example, be used as phenolic antioxidants:

(31) Alkylated monophenols, such as 2,6-di-tert-butyl-4-methylphenol, 2-tert-butyl-4,6-dimethylphenol, 2,6-di-tert-butyl-4-ethylphenol, 2,6-di-tert-butyl-4-n-butylphenol, 2,6-di-tert-butyl-4-isobutylphenol, 2,6-dicyclopentyl-4-methylphenol, 2-(α -methylcyclohexyl)-4,6-dimethylphenol, 2,6-dioctadecyl-4-methylphenol, 2,4,6-tricyclohexylphenol, 2,6-di-tert-butyl-4-methoxymethylphenol, linear or branched nonylphenols such as 2,6-dinonyl-4-methylphenol, 2,4-dimethyl-6-(1'-methylundec-1'-yl)phenol, 2,4-dimethyl-6-(1'-methylheptadec-1'-yl)phenol, 2,4-dimethyl-6-(1'-methyltridec-1'-yl)phenol and mixtures thereof; alkylthio methylphenols such as 2,4-dioctylthiomethyl-6-tert-butylphenol, 2,4-dioctylthiomethyl-6-methylphenol, 2,4-dioctylthiomethyl-6-ethylphenol, 2,6-didodecylthiomethyl-4-nonylphenol; hydroquinones and alkylated hydroquinones, such as 2,6-di-tert-butyl-4-methoxyphenol, 2,5-di-tert-butylhydroquinone, 2,5-di-tert-amylhydroquinone, 2,6-diphenyl-4-octadecyloxyphenol, 2,6-di-tert-butylhydroquinone, 2,5-di-tert-butyl-4-hydroxyanisole, 3,5-di-tert-butyl-4-hydroxyanisole, 3,5-di-tert-butyl-4-hydroxyphenylstearate, bis(3,5-di-tert-butyl-4-hydroxyphenyl) adipate; tocopherols such as α -, β -, γ -, δ -tocopherols and mixtures thereof (vitamin E); hydroxylated thiodiphenyl ethers such as 2,2'-thiobis(6-tert-butyl-4-methylphenol), 2,2'-thiobis(4-octylphenol), 4,4'-thiobis(6-tert-butyl-3-methylphenol), 4,4'-thiobis(6-tert-butyl-2-methylphenol), 4,4'-thiobis(3,6-di-sec-amylphenol), 4,4'-bis(2,6-dimethyl-4-hydroxyphenyl)disulfide; alkylidene bisphenols such as 2,2'-methylenebis(6-tert-butyl-4-methylphenol), 2,2'-methylenebis(6-tert-butyl-4-ethylphenol), 2,2'-methylenebis[4-methyl-6-(α -methylcyclohexyl)phenol], 2,2'-methylenebis(4-methyl-6-cyclohexylphenol), 2,2'-methylenebis(6-nonyl-4-methylphenol), 2,2'-methylenebis(4,6-di-tert-butylphenol), 2,2'-ethylidenebis(4,6-di-tert-butylphenol), 2,2'-ethylidenebis(6-tert-butyl-4-isobutylphenol), 2,2'-methylenebis[6-(α -methylbenzyl)-4-nonylphenol], 2,2'-methylenebis[6-(α , α -dimethylbenzyl)-4-nonylphenol], 4,4'-methylenebis-(2,6-di-tert-butylphenol), 4,4'-methylenebis(6-tert-butyl-2-methylphenol), 1,1-bis(5-tert-butyl-4-hydroxy-2-methylphenyl) butane, 2,6-bis(3-tert-butyl-5-methyl-2-hydroxybenzyl)-4-methylphenol, 1,1,3-tris(5-tert-butyl-4-hydroxy-2-methylphenyl) butane, 1,1-bis(5-tert-butyl-4-hydroxy-2-methylphenyl)-3-n-dodecylmercaptobutane, ethylene glycol-bis[3,3-bis(3'-tert-butyl-4'-hydroxyphenyl) butyrate], bis(3-tert-butyl-4-hydroxy-5-methylphenyl)dicyclopentadiene, bis[2-(3'-tert-butyl-2'-hydroxy-5'-methylbenzyl)-6-tert-butyl-4-methylphenyl]-terephthalate, 1,1-bis-(3,5-dimethyl-2-hydroxyphenyl) butane, 2,2-bis(3,5-di-tert-butyl-4-hydroxyphenyl) propane, 2,2-bis-(5-tert-butyl-4-hydroxy-2-methylphenyl)-4-n-dodecylmercaptobutane, 1,1,5,5-tetra(5-tert-butyl-4-hydroxy-2-methylphenyl) pentane O—, N- and S-benzyl compounds such as 3,5,3',5'-tetra-tert-butyl-4,4'-dihydroxydibenzylether, octadecyl-4-hydroxy-3,5-dimethylbenzylmercapto acetate, tridecyl-4-

hydroxy-3,5-di-tert-butylbenzylmercapto acetate, tris(3,5-di-tert-butyl-4-hydroxybenzyl)amine, bis(4-tert-butyl-3-hydroxy-2,6-dimethyl-benzyl)dithioterephthalate, bis(3,5-di-tert-butyl-4-hydroxybenzyl)sulfide, iso-octyl-3,5-di-tert-butyl-4-hydroxybenzylmercaptoacetate; hydroxybenzylated malonates such as dioctadecyl-2,2-bis(3,5-di-tert-butyl-2-hydroxybenzyl) malonate, dioctadecyl-2-(3-tert-butyl-4-hydroxy-5-methyl-benzyl) malonate, didodecylmercaptoethyl-2,2-bis(3,5-di-tert-butyl-4-hydroxy-benzyl) malonate, bis[4-(1,1,3,3-tetramethylbutyl)phenyl]--2,2-bis(3,5-di-tert-butyl-4-hydroxybenzyl) malonate; aromatic hydroxybenzyl compounds such as 1,3,5-tris(3,5-di-tert-butyl-4-hydroxybenzyl)-2,4,6-trimethylbenzol, 1,4-bis(3,5-di-tert-butyl-4-hydroxy-benzyl)-2,3,5,6-tetramethylbenzol, 2,4,6-tris(3,5-di-tert-butyl-4-hydroxy-benzyl)phenol; triazine compounds such as 2,4-bis(octylmercapto)-6-(3,5-di-tert-butyl-4-hydroxyanilino)-1,3,5-triazine, 2-octylmercapto-4,6-bis(3,5-di-tert-butyl-4-hydroxyanilino)-triazine, 2-octylmercapto-4,6-bis(3,5-di-tert-butyl-4-hydroxyphenoxy)-1,3,5-triazine, 2,4,6-tris(3,5-di-tert-butyl-4-hydroxyphenoxy)-1,2,3-triazine, 1,3,5-tris(3,5-di-tert-butyl-4-hydroxybenzyl) isocyanurate, 1,3,5-tris(4-tert-butyl-3-hydroxy-2,6-dimethylbenzyl) isocyanurate, 2,4,6-tris(3,5-di-tert-butyl-4-hydroxyphenylethyl)--triazine, 1,3,5-tris(3,5-di-tert-butyl-4-hydroxyphenylpropionyl) hexahydro--1,3,5-triazine, 1,3,5-tris(3,5-dicyclohexyl-4-hydroxybenzyl) isocyanurate; benzylphosphonates such as dimethyl-2,5-di-tert-butyl-4-hydroxybenzyl-phosphonate, diethyl-3,5-di-tert-butyl-4-hydroxybenzylphosphonate, diocta-decyl-3,5-di-tert-butyl-4-hydroxybenzylphosphonate, dioctadecyl-5-tert-butyl-4-hydroxy-3-methylbenzylphosphonate, the calcium salt of the monoethylester of the 3,5-di-tert-butyl-4-hydroxybenzyl phosphonic acid; acylaminophenols such as 4-hydroxylauranilide, 4-hydroxystearanilide, octyl-N-(3,5-di-tert-butyl-4-hydroxyphenyl) carbamate; esters of β -(3,5-di-tert-butyl-4-hydroxyphenyl) propionic acid with mono- or polyhydric alcohols, such as methanol, ethanol, n-octanol, i-octanol, octadecanol, 1,6-hexanediol, 1,9-nonanediol, ethylene glycol, 1,2-propanediol, neopentyl glycol, thiodiethylene glycol, diethylene glycol, triethylene glycol, pentaerythritol, tris(hydroxyethyl) isocyanurate, N,N'-bis(hydroxyethyl) oxamide, 3-thiaundecanol, 3-thiapentadecanol, trimethylhexanediol, trimethylolpropane, 4-hydroxymethyl-1-phospha-2,6,7-trioxabicyclo[2.2.2]octane; esters of the β -(5-tert-butyl-4-hydroxy-3-methylphenyl) propionic acid with monovalent or polyvalent alcohols such as methanol, ethanol, n-octanol, i-octanol, octadecanol, 1,6-hexanediol, 1,9-nonanediol, ethylene glycol, 1,2-propanediol, neopentyl glycol, thiodiethylene glycol, diethylene glycol, triethylene glycol, pentaerythritol, tris(hydroxyethyl) isocyanurate, N,N'-bis(hydroxyethyl) oxamide, 3-thiaundecanol, 3-thiapentadecanol, trimethylhexanediol, trimethylolpropane, 4-hydroxymethyl-1-phospha-2,6,7-trioxabicyclo[2.2.2]octane, 3,9-bis[2-{3-(3-tert-butyl-4-hydroxy-5-methylphenyl) propionyloxy}-1,1-dimethylethyl]-2,4,-2,4,8,10-tetra-oxaspiro[5.5]undecane; esters of the β -(3,5-dicyclohexyl-4-hydroxyphenyl) propionic acid with monovalent or polyvalent alcohols such as methanol, ethanol, octanol, octadecanol, 1,6-hexanediol, 1,9-nonanediol, ethylene glycol, 1,2-propanediol, neopentyl glycol, thiodiethylene glycol, diethylene glycol, triethylene glycol, pentaerythritol, tris(hydroxyethyl) isocyanurate, N,N'-bis(hydroxyethyl) oxamide, 3-thiaundecanol, 3-thiapentadecanol, trimethylhexanediol, trimethylolpropane, 4-hydroxymethyl-1-phospha-2,6,7-trioxabicyclo[2.2.2]octane; esters of the (3,5-di-tert-butyl-4-hydroxyphenyl) acetic acid with monovalent or polyvalent alcohols, e.g. methanol, ethanol, octanol, octadecanol, 1,6-hexanediol, 1,9-nonanediol, ethylene glycol, 1,2-propanediol, neopentyl glycol, thiodiethylene glycol, diethylene glycol, triethylene glycol, pentaerythritol, tris(hydroxyethyl) isocyanurate, N,N'-bis(hydroxyethyl) oxamide, 3-thiaundecanol, 3-thiapentadecanol, trimethylhexanediol, trimethylolpropane, 4-hydroxymethyl-1-phospha-2,6,7-trioxabicyclo[2.2.2]octane; amides of the β -(3,5-di-tert-butyl-4-hydroxyphenyl) propionic acid such as N,N'-bis(3,5-di-tert-butyl-4-hydroxyphenylpropionyl) hexamethylene diamide, N,N'-bis(3,5-di-tert-butyl-4-hydroxyphenylpropionyl) hexamethylene diamide, N,N'-bis(3,5-di-tert-butyl-4-hydroxyphenylpropionyl) hexamethylene diamide, N,N'-bis(3,5-di-tert-butyl-4-

hydroxyphenylpropionyl) hydrazide, N,N'-bis[2-(3-[3,5-di-tert-butyl-4-hydroxyphenyl]propionyloxy)ethyl]oxamide (Naugard®XL-1, marketed by Addivant); ascorbic acid (vitamin C).

(32) Particularly preferred phenolic antioxidants are the following structures:

(33) ##STR00003## ##STR00004## ##STR00005## ##STR00006## ##STR00007##
##STR00008##

where n in the compounds named above is a whole number and is between 2 and 10.

(34) Very particularly preferred phenolic antioxidants are octadecyl-3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate and pentaerythritol tetrakis(3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate).

(35) Further preferred phenolic antioxidants are phenolic antioxidants based on sustainable raw materials such as tocopherols (vitamin E), tocotrienols, tocomonoenols, ubichinols, hydroxytyrosol, flavanoids and flavanols such as chrysin, quercetin, hesperidin, neohesperidin, naningin, morin, kaempferol, fisein, datiscetin, luteolin, apigenin, taxifolin, isoflavones such as genistein, genistin, daidzein, daidzin, formononetin, anthocyanes such as delphinidin and malvidin, curcumin, carnosic acid, carnosol, rosmarinic acid, tannin and resveratrol, and carotenoids having alcoholic groups such as beta-cryptoxanthin, lutein, zeaxanthin, or astaxanthin.

(36) There can, for example, be used as aminic antioxidants:

(37) N,N'-di-isopropyl-p-phenylenediamine, N,N'-di-sec-butyl-p-phenylenediamine, N,N'-bis(1,4-dimethylpentyl)-p-phenylenediamine, N,N'-bis(1-ethyl-3-methyl-pentyl)-p-phenylenediamine, N,N'-bis(1-methylheptyl)-p-phenylenediamine, N,N'-dicyclohexyl-p-phenylenediamine, N,N'-diphenyl-p-phenylenediamine, N,N'-bis(2-naphthyl)-p-phenylenediamine, N-isopropyl-N'-phenyl-p-phenylenediamine, N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylene-diamine, N-(1-methylheptyl)-N'-phenyl-p-phenylenediamine, N-cyclohexyl-N'-phenyl-p-phenylenediamine, 4-(p-toluenesulfamoyl)diphenylamine, N,N'-dimethyl-N,N'-di-sec-butyl-p-phenylenediamine, diphenylamine, N-allyldiphenylamine, 4-isopropoxydiphenylamine, N-phenyl-1-naphthylamine, N-(4-tert-octylphenyl)-1-naphthylamine, N-phenyl-2-naphthylamine, octylated diphenylamine, e.g. p,p'-di-tert-octyldiphenylamine, 4-n-butylaminophenol, 4-butyrylamino-phenol, 4-nonanoylamino-phenol, 4-dodecanoylamino-phenol, 4-octadecanoylamino-phenol, bis(4-methoxyphenyl)amine, 2,6-di-tert-butyl-4-dimethylaminomethyl-phenol, 2,4'-di-aminodiphenyl methane, 4,4'-diaminodiphenyl methane, N,N,N',N'-tetra-methyl-4,4'-diaminodiphenylmethane, 1,2-bis[(2-methyl-phenyl)amino]ethane, 1,2-bis(phenylamino) propane, (o-tolyl) biguanide, bis[4-(1',3'-dimethylbutyl)phenyl]amine, tert-octylated N-phenyl-1-naphthylamine, a mixture of mono- and dialkylated tert-butyl/tert-octyldiphenylamines, a mixture of mono- and dialkylated nonyldiphenylamines, a mixture of mono- and dialkylated dodecyldiphenylamines, a mixture of mono- and dialkylated isopropyl/Isohexyl-diphenylamines, a mixture of mono- and dialkylated tert-butyl-diphenylamines, 2,3-dihydro-3,3-dimethyl-4H-1,4-benzothiazine, phenothiazine, a mixture of mono- and dialkylated tert-butyl/tert-octylphenothiazines, a mixture of mono- and dialkylated tert-octylphenothiazines, N-allylphenothiazine, N,N,N',N'-tetra-phenyl-1,4-diaminobut-2-ene and mixtures or combinations thereof.

(38) Preferred Aminic Antioxidants are: N,N'-di-isopropyl-p-phenylenediamine, N,N'-di-sec-butyl-p-phenylenediamine, N,N'-bis(1,4-dimethylpentyl)-p-phenylenediamine, N,N'-bis(1-ethyl-3-methylpentyl)-p-phenylenediamine, N,N'-bis(1-methylheptyl)-p-phenylenediamine, N,N'-dicyclohexyl-p-phenylenediamine, N,N'-diphenyl-p-phenylenediamine, N,N'-bis(2-naphthyl)-p-phenylenediamine, N-isopropyl-N'-phenyl-p-phenylenediamine, N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine, N-(1-methylheptyl)-N'-phenyl-p-phenylenediamine, N-cyclohexyl-N'-phenyl-p-phenylenediamine. Particularly preferred phenolic antioxidants are the following structures:

(39) ##STR00009## where n is a whole number and is between 3 and 100.

(40) Further preferred aminic antioxidants are: Hydroxylamines or N-oxides (nitrones) such as N,N-dialkylhydroxylamines, N,N-dibenzylhydroxylamine, N,N-dilaurylhydroxylamine, N,N-

distearylhydroxylamine, N-benzyl- α -phenylnitron, N-octadecyl- α -hexadecylnitron, and Genox EP (marketed by the SI Group) in accordance with the formula:

(41) ##STR00010##

(42) Preferred lactones are: Benzofuranones and indolinones such as 3-(4-(2-acetoxyethoxy)phenyl)-5,7-di-tert-butyl-benzofuran-2-one, 5,7-di-tert-butyl-3-[4-(2-stearoyloxyethoxy)phenyl]benzofuran-2-one, 3,3'-bis[5,7-di-tert-butyl-3-(4-(2-hydroxyethoxy)phenyl)benzofuran-2-one], 5,7-di-tert-butyl-3-(4-ethoxyphenyl)benzofuran-2-one, 3-(4-acetoxy-3,5-dimethylphenyl)-5,7-di-tert-butyl-benzofuran-2-one, 3-(3,5-dimethyl-4-pivaloyloxyphenyl)-5,7-di-tert-butyl-benzofuran-2-one, 3-(3,4-dimethylphenyl)-5,7-di-tert-butyl-benzofuran-2-one, 3-(2,3-dimethylphenyl)-5,7-di-tert-butyl-benzofuran-2-one as well as lactones that additionally contain a phosphite group such as

(43) ##STR00011##

A further suitable group of antioxidants are isoindolol[2,1-A]quinazolines such as

(44) ##STR00012##

(45) A further preferred variant of the method in accordance with the invention is characterized in that the at least one secondary antioxidant is selected from the group consisting of phosphorus compounds, in particular phosphites and phosphonites, organo-sulfur compounds, in particular sulfides and disulfides, and mixtures thereof.

(46) Suitable secondary antioxidants are in particular phosphites or phosphonites such as triphenylphosphite, diphenylalkylphosphites, phenyldialkylphosphites, tri (nonylphenyl)phosphite, trilaurylphosphites, trioctadecylphosphite, distearyl-pentaerythritoldiphosphite, tris-(2,4-di-tert-butylphenyl)phosphite, diisodecylpentaerythritoldiphosphite, bis(2,4-di-tert-butylphenyl) pentaerythritoldiphosphite, bis(2,4-di-cumylphenyl) pentaerythritoldiphosphite, bis(2,6-di-tert-butyl-4-methylphenyl)-pentaerythritoldiphosphite, diisodecyloxy-pentaerythritoldiphosphite, bis(2,4-di-tert-butyl-6-methylphenyl) pentaerythritoldiphosphite, bis(2,4,6-tris(tert-butylphenyl) penta-erythritoldiphosphite, tristearylsorbitoltriphosphite, tetrakis(2,4-di-tert-butylphenyl)-4,4'-biphenylendiphosphonite, 6-isooctyloxy-2,4,8, 10-tetra-tert-butyl-12H-dibenz[d,g]-1,3,2-dioxaphosphocin, bis(2,4-di-tert-butyl-6-methylphenyl)methylphosphite, bis(2,4-di-tert-butyl-6-methyl-phenyl)-ethylphosphite, 6-fluoro-2,4,8,10-tetra-tert-butyl-12-methyl-dibenz-[d,g]-,2-1,3dioxaphosphocin, 2,2'2''-nitrilo[triethyltris(3,3'',5,5'-tetra-tert-butyl-1,1'-biphenyl-2,2'-diyl)phosphite], 2-ethylhexyl(3,3',5,5'-tetra-tert-butyl-1,1'-biphenyl-2,2'-diyl))phosphite, 5-butyl-5-ethyl-2-(2,4,6-tri-tert-butyl-phenoxy)-1,3,2-dioxaphosphirane.

(47) Particularly preferred phosphites/phosphonites are:

(48) ##STR00013## ##STR00014## ##STR00015##

where n=3-100.

(49) A preferred phosphonite is:

(50) ##STR00016##

(51) The phosphite tris-(2,4-di-tert-butylphenyl)phosphite is very particularly preferably used as the secondary antioxidant.

(52) Suitable secondary antioxidants are furthermore organosulfur compounds such as sulfides and disulfides, e.g. distearylthiodipropionate, dilaurylthiodipropionate; ditridecylthiodipropionate, ditetradecylthiodipropionate, 3-(dodecylthio)-,1,1'-[2,2-bis[[3-(dodecylthio)-1-oxopropoxy]methyl]-1,3-propandiyl]propanoic acid ester. The following structures are preferred:

(53) ##STR00017##

(54) In a further preferred embodiment, component (B) comprises both a primary antioxidant and a secondary antioxidant. It is particularly preferred in this case that component (B) comprises a phenolic antioxidant and a phosphite or phosphonite.

(55) In a further preferred embodiment, components (A) and (B) comprise a sustainable raw material.

(56) A particularly preferred variant of the method in accordance with the invention is characterized

in that at least one primary antioxidant and at least one secondary antioxidant are introduced into the thermoplastic. This means that at least one primary antioxidant and at least one secondary antioxidant are used as component (B). In this manner, an even more effective stabilization of the thermoplastic against oxidative, thermal, and/or actinic degradation can be achieved.

(57) In a further preferred variant of the method in accordance with the invention, component (A) and component (B) are introduced into the thermoplastic plastic recycle in a weight ratio of 5:95 to 95:5, preferably of 10:90 to 90:10, particularly preferably of 20:80 to 80:20.

(58) If component (B) comprises both a primary antioxidant and a secondary antioxidant, the ratio between the primary antioxidant and the secondary antioxidant is preferably 90:10 to 10:90, particularly preferably 80:20 to 20:80, and very particularly preferably 70:30 to 30:70.

(59) A further preferred variant of the method in accordance with the invention is characterized in that components (A) and (B) are introduced into the thermoplastic plastic recycle in that components (A) and/or (B) present as solids are mixed with the thermoplastic plastic recycle present as a solid and the mixture thereby produced is melted and subsequently cooled; or components (A) and/or (B) present as solids are melted and the melt thereby produced is introduced into a melt of the thermoplastic plastic recycle.

(60) Components (A) and (B), that can be present as a powder, compacted, as pellets, a solution, or flakes, are preferably mixed with the polymer to be stabilized, the polymer matrix is transferred into the melt and is subsequently cooled. Alternatively to this, it is equally possible to introduce the additive into a polymer melt in a molten state.

(61) For the case that further components are added to the thermoplastic plastic recycle, they can be admixed to the polymers separately, in the form of liquids, powders, pellets, or compacted products, or together with the additive composition in accordance with the invention as described above.

(62) At least one additive can preferably additionally be introduced into the thermoplastic plastic recycle that is selected from the group consisting of UV absorbers, light stabilizers, metal deactivators, filler deactivators, antiozonants, nucleation agents, anti-nucleation agents, impact strength enhancers, plasticizers, lubricants, rheology modifiers, thixotropic agents, chain extenders, optical brighteners, antimicrobial active agents, antistatic agents, slip agents, anti-blocking means, coupling agents, crosslinking agents, anti-cross-linking agents, hydrophilization agents, hydrophobing agents, bonding agents, dispersing agents, compatibilizers, oxygen scavengers, acid scavengers, expanding agents, degradation additives, defoaming agents, odor scavengers, marking agents, anti-fogging agents, fillers, reinforcement agents, and mixtures thereof.

(63) It is further preferred that at least one additive is additionally introduced into the thermoplastic plastic recycle that is selected from the group consisting of a) acid scavengers, light stabilizers, preferably light stabilizers from the group of hindered amines, b) dispersing agents, c) filler deactivators, and mixtures thereof.

(64) Suitable acid scavengers ("antacids") are salts of monovalent, bivalent, trivalent, or quadrivalent metals, preferably alkali metals, alkaline earth metals, aluminum or zinc, in particular formed with fatty acids such as calcium stearate, magnesium stearate, zinc stearate, aluminum stearate, calcium laurate, calcium behenate, calcium lactate, calcium stearyl-2-lactate. Further classes of suitable acid scavengers are hydrolactites, in particular synthetic hydrolactites on the basis of aluminum, magnesium and zinc, hydrocalumites, zeolites, alkaline earth oxides, in particular calcium oxide and magnesium oxide and zinc oxide, alkaline earth carbonates, in particular calcium carbonate, magnesium carbonate, dolomite, and hydroxides, in particular brucite (magnesium hydroxide).

(65) Suitable light stabilizers are, for example, compounds based on 2-(2'-hydroxyphenyl)benzotriazoles, 2-hydroxy benzophenones, esters of benzoic acids, acrylates, oxamides, and 2-(2-hydroxyphenyl)-1,3,5-triazines.

(66) Suitable 2-(2'-hydroxyphenyl)benzotriazoles are, for example, 2-(2'-hydroxy-5'-

methylphenyl)benzotriazole, 2-(3',5'-di-tert-butyl-2'-hydroxy-phenyl)-benzotriazole, 2-(5'-tert-butyl-2'-hydroxy-phenyl)benzotriazole, 2-(2'-hydroxy-5'-(1,1,3,3-tetramethylbutyl)phenyl)benzotriazole, 2-(3',5'-di-tert-butyl-2'-hydroxyphenyl)-5-chlorobenzotriazole, 2-(3'-tert-butyl-2'-hydroxy-5'-methylphenyl-5-chlorobenzotriazole, 2-(31-sec-butyl-5'-tert-butyl-2'-hydroxy-phenyl)benzotriazole, 2-(2'-hydroxy-4'-octyloxyphenyl)benzotriazole, 2-(31,5'-di-tert-amyl-2'-hydroxyphenyl)benzotriazole, 2-(3',5'-bis(α,α -dimethylbenzyl)-2'-hydroxyphenyl)benzotriazole, 2-(3'-tert-butyl-2'-hydroxy-5'-(2-octyloxy-carbonylethyl)phenyl)-5-chlorobenzotriazole, 2-(3'-tert-butyl-5'-[2-(2-ethyl-hexyloxy)carbonylethyl]-2'-hydroxyphenyl)-5-chlorobenzotriazole, 2-(3'-tert-butyl-2'-hydroxy-5'-(2-methoxycarbonylethyl)phenyl)-5-chlorobenzotriazole, 2-(3'-tert-butyl-2'-hydroxy-5'-(2-methoxycarbonylethyl)phenyl)benzotriazole, 2-(3'-tert-butyl-2'-hydroxy-5'-(2-octyloxy-carbonylethyl)phenyl)benzotriazole, 2-(3'-tert-butyl-5'-[2-(2-ethylhexyloxy)carbonylethyl]-2'-hydroxyphenyl)benzotriazole, 2-(3'-dodecyl-2'-hydroxy-5'-methylphenyl)benzotriazole, 2-(3'-tert-butyl-2'-hydroxy-5'-(2-isooctyloxy-carbonylethyl)phenyl)benzotriazole, 2,2'-methylenebis[4-(1,1,3,3-tetramethylbutyl)-6-benzotriazol-2-ylphenol]; the product of the transesterification of 2-[3'-tert-butyl-5'-(2-methoxycarbonylethyl)-2'-hydroxyphenyl]-2H-benzotriazole with polyethylene glycol 300; [R—CH₂CH₂—COO—CH₂CH₂—]₂, where R=3'-tert-butyl-4'-hydroxy-5'-2H-benzotriazol-2-ylphenyl, 2-[2'-hydroxy-3'-(α,α -dimethylbenzyl)-5'-(1,1,3,3-tetramethylbutyl)-phenyl]benzotriazole, 2-[2'-hydroxy-3'-(1,1,3,3-tetramethylbutyl)-5'-(α,α -dimethylbenzyl)phenyl]benzotriazole.

(67) Suitable 2-hydroxybenzophenones are, for example, 4-hydroxy-, 4-methoxy-, 4-octyloxy-, 4-decyloxy-4-dodecyloxy, 4-benzyloxy, 2',4'-trihydroxy- and 2'-hydroxy-4,4'-dimethoxy derivatives of the 2-hydroxy benzophenones.

(68) Suitable acrylates are, for example, ethyl- α -cyano- β , β -diphenylacrylate, isooctyl- α -cyano- β , β -diphenylacrylate, methyl- α -carbomethoxycinnamate, methyl- α -cyano- β -methyl-p-methoxycinnamate, butyl- α -cyano- β -methyl-p-methoxycinnamate, methyl- α -carbomethoxy-p-methoxycinnamate and N—(β -carbomethoxy- β -cyanovinyl)-2-methylindoline.

(69) Suitable esters of benzoic acid are, for example 4-tert-butylphenyl salicylate, phenylsalicylate, octylphenylsalicylate, dibenzoyl resorcinol, bis(4-tert-butylbenzoyl) resorcinol, benzoyl resorcinol, 2,4-di-tert-butylphenyl-3,5-di-tert-butyl-4-hydroxybenzoate, hexadecyl-3,5-di-tert-butyl-4-hydroxybenzoate, octadecyl-3,5-di-tert-butyl-4-hydroxybenzoate, 2-methyl-4,6-di-tert-butylphenyl-3,5-di-tert-butyl-4-hydroxybenzoate.

(70) Suitable oxamides are, for example, 4,4'-dioctyloxyoxanilide, 2,2'-diethoxyoxanilide, 2,2'-dioctyloxy-5,5'-di-tert-butoxanilide, 2,2'-didodecyloxy-5,5'-di-tert-butoxanilide, 2-ethoxy-2'-ethyloxanilide, N,N'-bis(3-dimethylaminopropyl) oxamide, 2-ethoxy-5-tert-butyl-2'-ethoxanilide and its mixture with 2-ethoxy-2'-ethyl-5,4'-di-tert-butoxanilide, mixtures of o- and p-methoxy-disubstituted oxanilides and mixtures of o- and p-ethoxy-disubstituted oxanilides.

(71) Suitable 2-(2-hydroxyphenyl)-1,3,5-triazines are, for example, 2,4,6-tris(2-hydroxy-4-octyloxyphenyl)-1,3,5-triazine, 2-(2-hydroxy-4-octyloxyphenyl)-4,6-bis(2,4-dimethylphenyl)-1,3,5-triazine, 2-(2,4-dihydroxyphenyl)-4,6-bis(2,4-dimethylphenyl)-1,3,5-triazine, 2,4-bis(2-hydroxy-4-propyloxyphenyl)-6-(2,4-dimethylphenyl)-1,3,5-triazine, 2-(2-hydroxy-4-octyloxyphenyl)-4,6-bis(4-methylphenyl)-1,3,5-triazine, 2-(2-hydroxy-4-dodecyloxyphenyl)-4,6-bis(2,4-dimethylphenyl)-1,3,5-triazine, 2-(2-hydroxy-4-tridecyloxyphenyl)-4,6-bis(2,4-dimethylphenyl)-1,3,5-triazine, 2-[2-hydroxy-4-(2-hydroxy-3-butyloxypropoxy)-phenyl]-4,6-bis(2,4-dimethyl)-1,3,5-triazine, 2-[2-hydroxy-4-(2-hydroxy-3-octyloxypropoxy)phenyl]-4,6-bis(2,4-dimethyl)-1,3,5-triazine, 2-[4-(dodecyloxy-/tridecyloxy-2-hydroxypropoxy)-2-hydroxyphenyl]-4,6-bis(2,4-dimethylphenyl)-1,3,5-triazine, 2-[2-hydroxy-4-(2-hydroxy-3-dodecyloxypropoxy)phenyl]-4,6-bis(2,4-dimethylphenyl)-1,3,5-triazine, 2-(2-hydroxy-4-hexyloxy)phenyl-4,6-diphenyl-1,3,5-triazine, 2-(2-hydroxy-4-methoxyphenyl)-4,6-

diphenyl-1,3,5-triazine, 2,4,6-tris[2-hydroxy-4-(3-butoxy-2-hydroxypropoxy)phenyl]-1,3,5-triazine, 2-(2-hydroxyphenyl)-4-(4-methoxyphenyl)-6-phenyl-1,3,5-triazine, 2-{2-hydroxy-4-[3-(2-ethylhexyl-1-oxy)-2-hydroxypropyloxy]phenyl}-4,6-bis(2,4-dimethylphenyl-1,3,5-triazine).
(72) Suitable metal deactivators are, for example, N,N'-diphenyloxamide, N-salicylale-N'-salicyloylhydrazine, N,N'-bis(salicyloyl) hydrazine, N,N'-bis(3,5-di-tert-butyl-4-hydroxyphenylpropionyl) hydrazine, 3-salicyloylamino--triazole, bis(benzylidene) oxalyldihydrazide, oxanilide, isophthaloyldihydrazide, sebacoyl-bis-phenylhydrazide, N,N'-diacetyladi-poyldihydrazide, N,N'-bis(salicyloyl)-oxylyldihydrazide, N,N'-bis(salicyloyl)thiopropionylhydrazide.

(73) Particularly preferred as metal deactivators are

(74) ##STR00018##

(75) Suitable hindered amines are, for example, 1,1-bis(2,2,6,6-tetramethyl-4-piperidyl)succinate, bis(1,2,2,6,6-pentamethyl-4-piperidyl)sebazate, bis(1-octyloxy-2,2,6,6-tetramethyl-4-piperidyl)sebazate, bis(1, 2,2,6, 6-pentamethyl-4-piperidyl)-n-butyl-3,5-di-tert-butyl-4-hydroxybenzylmalonate, the condensation product of 1-(2-hydroxyethyl)-2,2,6,6-tetramethyl-4-hydroxy-piperidine und succinic acid, linear or cyclic condensation products of N,N'-bis(2,2,6,6-tetramethyl-4-piperidyl) hexamethylenediamine and 4-tert-octylamino-2,6-di-chloro-1,3,5-triazine, tris(2,2,6,6-tetramethyl-4-piperidyl) nitrilotriacetate, tetrakis(2,2,6,6-tetra-methyl-4-piperidyl)-1,2,3,4-butantetracarboxylate, 1,1'-(1,2-ethandiyl)-bis(3,3,5,5-tetramethylpiperazinone), 4-benzoyl-2,2,6,6-tetramethylpiperidine, 4-stearyloxy-2, 2,6,6-tetramethylpiperidine, linear or cyclic condensation products of N,N'-bis(2,2,6,6-tetramethyl-4-piperidyl) hexamethylenediamine and 4-morpholino-2,6-dichloro-1,3,5-triazine, the reaction product of 7, 7, 9, 9-tetramethyl-2-cycloundecyl-1-oxa-3,8-diaza-4-oxospiro-[4,5]decane and epichlorohydrin.

(76) The sterically hindered N—H, N-alkyl such as N-methyl or N-octyl, the N-alkoxy derivatives such as N-methoxy or N-octyloxy, the cycloalkyl derivatives such as N-cyclohexyloxy and the N-(2-hydroxy-2-methylpropoxy) analogs are also each included in the above-given structures here.

(77) Preferred hindered amines furthermore have the following structures:

(78) ##STR00019## ##STR00020## ##STR00021##

(79) Preferred oligomeric and polymeric hindered amines have the following structures:

(80) ##STR00022## ##STR00023##

(81) In the above-named compounds, n respectively means 3 to 100.

(82) A further suitable light stabilizer is Hostanox NOW (manufacturer: Clariant SE) having the following general structure:

(83) ##STR00024##

(84) Suitable dispersion agents are, for example: polyacrylates, e.g. copolymers with long-chain side groups, polyacrylate block copolymers, alkylamides: e.g. N,N'—1,2-ethanediylbisoctadecanamide sorbitan esters, e.g. monostearylsorbitan esters, titanates and zirconates, reactive copolymers with functional groups, e.g. polypropylene-co-acrylic acid, polypropylene-co-maleic acid anhydride, polyethylene-co-glycidylmethacrylate, polystyrene-alt-maleic acid anhydride polysiloxanes: e.g. dimethylsilanediol-ethylene oxide copolymers, polyphenylsiloxane copolymers, amphiphilic copolymers: e.g. polyethylene block polyethylene oxide, dendrimers, e.g. dendrimers containing hydroxyl groups.

(85) Suitable antinucleation agents are azine dyes such as nigrosin.

(86) Suitable flame retardant agents are, in particular a) inorganic flame retardant agents such as Al(OH).sub.3, Mg(OH).sub.2, AlO(OH), MgCO.sub.3, sheet silicates such as montmorillonite or sepiolite, non-modified or organically modified double salts such as Mg—Al silicates, POSS (polyhedral oligomeric silsesquioxane) compounds, huntite, hydro magnesite or halloysite and Sb.sub.2H.sub.3, Sb.sub.2H.sub.5, MoO.sub.3, zinc stannate, zinc hydroxystannate, b) flame retardant agents containing nitrogen such as melamine, melem, melam, melon, melamine derivatives, melamine condensation products or melamine salts, benzoguanamine,

polyisocyanurates, allantoin, phosphacenes, in particular melamine cyanurate, melamine phosphate, dimelamine phosphate, melamine pyrophosphate, melamine polyphosphate, melamine metal phosphates such as melamine aluminum phosphate, melamine zinc phosphate, melamine magnesium phosphate, and the corresponding pyrophosphates and polyphosphates, poly-[2,4-(piperazine-1,4-yl)-6-(morpholine-4-yl)-triazine], ammonium polyphosphate, melamine borate, melamine hydrobromide, c) radical formers such as alkoxyamines, hydroxylamine esters, azo compounds, triazine compounds, disulfides, polysulfides, thiols, thiuram sulfides, dithiocarbamates, mercaptobenzthiazoles, sulfene amides, sulfene imides, d) dicumyl or polycumyl, hydroxyimides and their derivatives such as hydroxyimide esters or hydroxyimide ethers, e) flame retardant agents containing phosphorus such as red phosphorus, phosphates such as resorcin diphosphite, bisphenol A diphosphate, and their oligomers, triphenylphosphate, ethylene diamine diphosphate, phosphinates such as salts of the hypophosphorous acid and their derivatives such as alkylphosphinate salts, e.g. diethylphosphinate aluminum or diethylphosphinate zinc or aluminum phosphinate, aluminum phosphite, aluminum phosphonate, phosphonate esters, oligomer and polymer derivatives of the methane phosphonic acid, 9,10-dihydro-9-oxa-10-phosphorylphenanthrene-10-oxide (DOPO) and their substituted compounds, f) halogenated flame retardant agents based on chlorine and bromine such as polybrominated diphenyl oxides such as decabromodiphenyl oxide, tris(3-bromo-2,2-bis(bromomethyl) propylphosphate, tris(tribromoneopentyl)-phosphate, tetrabromophthalic acid, 1,2-bis(tribromophenoxy) ethane, hexabromocyclododecane, brominated diphenylethane, tris-(2,3-dibromopropyl) isocyanurate, ethylene-bis-(tetrabromophthalimide), tetrabromo-bisphenol A, brominated polystyrene, brominated polybutadiene or polystyrene brominated polybutadiene copolymers, brominated polyphenylene ether, brominated epoxy resin, polypentabromobenzylacrylate, optionally in combination with Sb.sub.2H.sub.3 and/or Sb.sub.2H.sub.5, g) borates such as zinc borate or calcium borate, optionally on a carrier material such as silica. h) sulfurous compounds, such as elemental sulfur, disulfides and polysulfides, thiuram sulfide, dithiocarbamates, mercaptobenzothiazole and sulfene amides, i) anti-drip agents, such as polytetrafluoroethylene, j) silicon-containing compounds, such as polyphenyl siloxanes, k) carbon modifications, such as carbon nanotubes (CNTs), expandable graphite or graphene l) and combinations or mixtures thereof.

(87) The following compounds are halogen free and are very particularly preferred flame retardant agents: Al(OH).sub.3, Mg(OH).sub.2,

(88) ##STR00025## ##STR00026##

with R=alkyl, phenyl respectively and n=3 to 20

(89) ##STR00027##

(90) Suitable plasticizers are, for example, phthalic acid esters, adipic acid esters, esters of citric acid, ester of 1,2-cyclohexane dicarboxylic acid, trimellitic acid esters, isorobide esters, phosphate esters, epoxides such as epoxidized soy bean oil, or aliphatic polyesters. Suitable mold lubricants and processing aids are, for example, polyethylene waxes, polypropylene waxes, salts of fatty acids such as calcium stearate, zinc stearate, or salts of montane waxes, amide waxes such as erucic acid amide or oleic acid amides, fluoropolymers, silicones, or neoalkoxytitanates and zirconates.

(91) Suitable pigments can be of an inorganic or organic nature. Inorganic pigments are, for example, titanium dioxide, zinc oxide, zinc sulfide, iron oxide, ultramarine, black carbon; organic pigments are, for example, anthraquinones, anthanthrones, benzimidazolones, chinacridones, diketopyrrolopyrrols, dioxazines, inanthrones, isoindolines, azo compounds, perylenes, phthalocyanines or pyranthrones. Further suitable pigments include effect pigments on a metal basis or pearl gloss pigments on a metal oxide basis.

(92) Suitable optical brighteners are, for example, bis-beznzoxazoles, phenylcumarines, or bis(styryl) biphenyls and in particular optical brighteners of the formulas:

(93) ##STR00028##

(94) Suitable filler deactivators are, for example, polysiloxanes, polyacrylates, in particular block copolymers such as polymethacrylic acid polyalkylene oxide or polyglycidyl(meth)acrylates and their copolymers, e.g. with styrene and epoxides of e.g. the following structures:

(95) ##STR00029##

(96) Suitable antistatic agents are, for example, ethoxylated alkylamines, fatty acid esters, alkylsulfonates, and polymers such as polyetheramides.

(97) Suitable antiozonants are the above-named amines such as N,N'-di-isopropyl-p-phenylenediamine, N,N'-di-sec-butyl-p-phenylenediamine, N,N'-bis(1,4-dimethylpentyl)-p-phenylenediamine, N,N'-dicyclohexyl-p-phenylenediamine, N-isopropyl-N'-phenyl-p-phenylenediamine, N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine, N-(1-methylheptyl)-N'-phenyl-p-phenylenediamine, N-cyclohexyl-N'-phenyl-p-phenylenediamine. Suitable rheology modifiers, e.g. for the preparation of controlled rheology polypropylene (CR-PP), are, for example, peroxides, alkoxyaminoesters, oxymide sulfonic acid esters, and in particular the following structures:

(98) ##STR00030##

(99) Suitable nucleation agents are, talcum, alkali, or alkaline earth salts of mono- and polyfunctional carboxylic acids such as benzoic acid, succinic acid, adipic acid, e.g. sodium benzoate, zinc glycerolate, aluminiumhydroxy-bis(4-tert-butyl)benzoate, 2,2'-methylene-bis-(4,6-di-tert-butylphenyl)phosphate, and trisamides and diamides such as trimesic acid tricyclohexylamide, trimesic acid tri (4-methylcyclohexylamide), trimesic acid tri (tert-butylamide), N,N',N''-1,3,5 benzotriyltris(2,2-dimethyl-propanamide) or 2,6-naphthalene dicarboxylic acid cyclohexylamide.

(100) Suitable additives for the linear molecular weight makeup of polycondensation polymers (chain extenders) are diepoxides, bis-oxazonines, bis-oxazolones, bis-oxazines, diisocyanates, dianhydrides, bis-acyllactams, bis-maleimides, dicyanates, carbodiimides. Further suitable chain extenders include polymer compounds such as polystyrene polyacrylate polyglycidyl(meth)acrylate copolymers, polystyrene maleic acid anhydride copolymers, and polyethylene maleic acid anhydride copolymers.

(101) Suitable additives to increase the electrical conductivity are, for example, the mentioned static inhibitors, black carbon, and carbon compounds such as carbon nanotubes and graphene, metal powders such as copper powder, and conductive polymers such as polypyrroles, polyanilines, and polythiopenes. Suitable additives to increase thermal conductivity are, for example, aluminum nitrides and boron nitrides.

(102) Suitable infrared-active additives are, for example, aluminum silicates or dyestuffs such as phthalocyanines or anthraquinones.

(103) Suitable demolding agents are, for example, silicones, soaps, and waxes, such as montan waxes.

(104) The incorporation of components (A) and (B) and optionally of the additional additives into the plastic recyclate takes place by typical processing methods, with the polymers being melted and being mixed with the additive composition in accordance with the invention and the optional further additives, preferably by mixers, kneaders and extruders. Extruders, such as single-screw extruders, twin-screw extruders, planetary gear extruders, ring extruders, co-kneaders, are preferred processing machines and are preferably equipped with a vacuum degassing means. The processing can here take place under air or optionally under inert gas conditions such as under nitrogen.

(105) Components (A) and (B) can furthermore be manufactured and introduced in the form of so-called master batches or concentrates that, for example, comprise 10-90% of a stabilizer composition comprising components (A) and (B) in a polymer.

(106) It is further preferred that the thermoplastic plastic recyclate is selected from the group consisting of a) recycled polymers of olefins or diolefins such as polyethylene, in particular LDPE, LLDPE, VLDPE, ULDPE, MDPE, HDPE, and UHMWPE, metallocene PE (mPE), polypropylene,

polyisobutylene, poly-4-methyl-pentene-1, polybutadiene, polyisoprene, polycyclooctene, polyalkylene carbon monoxide copolymers, and corresponding copolymers in the form of statistical or block structures such as polypropylene-polyethylene (EP), EPM or EPDM, ethylene vinyl acetate (EVA), ethylene acrylic ester such as ethylene butyl acrylate, ethylene acrylic acid glycidyl acrylate, and corresponding graft polymers such as polypropylene-g-maleic acid anhydride, polypropylene-g-acrylic acid, and polyethylene-g-acrylic acid. b) recycled polystyrene, polymethyl styrene, polyvinyl naphthalene, styrene butadiene (SB), styrene butadiene styrene (SBS), styrene ethylene butylene styrene (SEBS), styrene ethylene propylene styrene, styrene isoprene, styrene isoprene styrene (SIS), styrene butadiene acrylonitrile (ABS), styrene acrylonitrile acrylate (ASA), styrene ethylene, styrene maleic acid anhydride polymers, including corresponding graft copolymers such as styrene on butadiene, maleic acid anhydride on SBS or SEBS, and graft copolymers of methyl methacrylate, styrene butadiene, and ABS (MABS), c) recycled polymers of unsaturated esters such as polyacrylates and polymethacrylates such as polymethyl methacrylate (PMMA), polybutyl acrylate, polyacryl acrylate, poly stearyl acrylate, polyacrylonitrile, and corresponding polyacrylamides, copolymers such as polyacrylonitrile-polyalkylacrylate, d) recycled polymers of unsaturated alcohols and derivatives such as polyvinyl alcohol, polyvinyl acetate, polyvinyl butyral, e) recycled polyacetates such as polyoxymethylene (POM), and corresponding copolymers such as copolymers with butanal, f) recycled polyphenylene oxides and blends with polystyrene or polyamides, g) recycled polymers of cyclic ethers such as polyethylene glycol, polypropylene glycol, polyethylene oxide, polypropylene oxide, h) recycled polyurethanes of hydroxy terminated polyethers or polyesters and aromatic or aliphatic isocyanates, in particular linear polyurethanes, polyureas, i) recycled polyamides such as polyamide-6, 6.6, 6.10, 4.6, 4.10, 6.12, 12.12, polyamide 11, polyamide 12 and (partly) aromatic polyamides such as polyphthalamides, e.g. prepared from terephthalic acid and/or isophthalic acid and aliphatic diamines or from aliphatic dicarboxylic acids such as adipic acid or sebacic acid and aromatic diamines such as 1,4- or 1,3-diaminobenzol, j) recycled polyimides, polyamide imides, polyether imides, polyester imides, poly(ether)ketones, polysulfones, polyether sulfones, polyaryl sulfones, polyphenylene sulfides, polybenzimidazole azoles, polyhydantoin, k) recycled polyesters of aliphatic or aromatic dicarboxylic acids and diols or of hydroxy carboxylic acids such as polyethylene terephthalate (PET), polybutylene terephthalate (PBT), polypropylene terephthalate, polyethylene naphthylate, poly-1,4-dimethylcyclohexane terephthalate, polyhydroxy benzoate, polyhydroxy naphthylate, polylactic acid, l) recycled polycarbonates, polyester carbonates, and blends thereof such as PC/ABS, PC/PBT, PC/PET/PBT, m) recycled halogenated polymers such as PVC, PVDC, n) recycled cellulose derivatives such as cellulose nitrate, cellulose acetate, cellulose propionate, cellulose butyrate, o) and mixtures, combinations, or blends of two or more of the above-named polymers.

(107) The thermoplastic plastic recyclate is particularly preferably selected from the group consisting of polymers of olefins or diolefins such as polyethylene, in particular LDPE, LLDPE, VLDPE, ULDPE, MDE, HDPE, and UHMWPE, metallocene PE (m-PE), polypropylene, polyisobutylene, poly-4-methyl-pentene-1, polybutadiene, polyisoprene, polycyclooctene, polyalkylene carbon monoxide copolymers, and corresponding copolymers in the form of statistical or block structures such as polypropylene-polyethylene (EP), EPM or EPDM, ethylene vinyl acetate (EVA), ethylene-acrylic esters such as ethylene butyl acrylate, ethylene acrylic acid glycidyl acrylate, and corresponding graft polymers such as polypropylene-g-maleic acid anhydride, polypropylene-g-acrylic acid, and polyethylene acrylic acid.

(108) The thermoplastic is very particularly preferably a polyolefin recyclate. The thermoplastic plastic recyclate can, for example, be polypropylene, that is in particular a polypropylene recyclate or polyethylene, that is in particular a polyethylene recyclate.

(109) In the method in accordance with the invention, (A).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one substituted sugar, and (B).sub.0.04 to 6 parts by weight,

preferably 0.06 to 1 parts by weight, of at least one primary antioxidant and/or at least one secondary antioxidant

are furthermore preferably introduced into 91 to 99.94 parts by weight, preferably 98.5 to 99.91 parts by weight, of at least one thermoplastic plastic recycle.

(110) It is further preferred that (A).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one substituted sugar, (B1).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one primary antioxidant, (B2).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one secondary antioxidant

are introduced into 91 to 99.94 parts by weight, preferably 98.5 to 99.91 parts by weight, of at least one thermoplastic plastic recycle.

(111) The present invention further relates to a plastic composition comprising or consisting of (A) at least one substituted sugar (component (A)), and (B) at least one thermoplastic plastic recycle.

(112) In this case—with the exception of the primary and/or secondary antioxidants already contained in the recycle—no further primary and/or secondary (fresh) antioxidants are admixed in the plastic composition.

(113) The primary and/or secondary antioxidants contained in the recycle and their degradation products can, where required, be determined by conventional analytical methods familiar to the skilled person, e.g. by chromatographic methods such as HPLC, optionally after extraction from the polymers by means of a suitable solvent, and subsequent concentration by removal of the solvent.

(114) The present invention additionally also relates to a plastic composition comprising or consisting of (A) at least one substituted sugar, (B) at least one primary antioxidant and/or at least one secondary antioxidant; and (C) at least one thermoplastic plastic recycle.

(115) In a preferred embodiment, the plastic composition is characterized in that the plastic composition comprises or consists of (A).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one substituted sugar; (B).sub.0.04 to 6 parts by weight, preferably 0.06 to 1 parts by weight, of at least one primary antioxidant and/or of at least one secondary antioxidant, and (C).sub.91 to 99.94 parts by weight, preferably 98.5 to 99.91 parts by weight of at least one thermoplastic.

(116) The composition in accordance with the invention preferably comprises at least one primary antioxidant and at least one secondary antioxidant.

(117) A further preferred embodiment is characterized in that the plastic composition comprises or consists of (A).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one substituted sugar; (B1).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one primary antioxidant; (B2).sub.0.02 to 3 parts by weight, preferably 0.03 to 0.5 parts by weight, of at least one secondary antioxidant, and (C).sub.91 to 99.94 parts by weight, preferably 98.5 to 99.91 parts by weight of at least one thermoplastic plastic recycle.

(118) In accordance with a further preferred embodiment the plastics composition according to the invention additionally contains at least one additive which is selected from the group consisting of UV absorbers, light stabilizers, metal deactivators, filler deactivators, antiozonants, nucleation agents, antinucleation agents, impact modifiers, plasticizers, lubricants, rheology modifiers, thixotropy agents, chain extenders, optical brighteners, antimicrobial active substances, antistatic agents, slip agents, antiblocking agents, coupling means, crosslinking agents, anticrosslinking agents, hydrophilizing agents, hydrophobing agents, adhesion promoters, dispersants, compatibilizers, oxygen scavengers, acid scavengers, propellants, degradation additives, antifoaming agents, odor scavengers, markers, antifogging agents, fillers, reinforcements, and mixtures hereof.

(119) It is furthermore preferred that the plastics composition additionally contains at least one additive which is selected from the group consisting of a) acid scavengers, preferably calcium stearate, magnesium stearate, zinc stearate, aluminum stearate, calcium stearate, calcium lactate,

calcium stearoyl-2-lactate, hydrotalcites, in particular synthetic hydrotalcites based on aluminum, magnesium and zinc, hydrocalumites, zeolites, alkaline earth oxides, in particular calcium oxide and magnesium oxide, zinc oxide, alkaline earth carbonates, in particular calcium carbonate, magnesium carbonate and dolomite, and hydroxides, in particular brucite, b) light stabilizers, preferably light stabilizers from the group of hindered amines, c) dispersants, d) filler deactivators, and mixtures thereof.

(120) If only a substituted sugar is added to the recyclate as a stabilizer, the further additive is in particular an acid scavenger in a preferred embodiment. Preferred acid scavengers are defined above.

(121) The plastic composition in accordance with the invention can preferably be manufactured or has been manufactured using a method in accordance with the invention for the stabilization of thermoplastics against oxidative, thermal, and/or actinic degradation.

(122) All the preferred variants, exemplary embodiments, and comments (e.g. with respect to possible components and additives to be used) described with respect to the method in accordance with the invention also apply accordingly to the plastic composition in accordance with the invention.

(123) The present invention also relates to a molding compound or to a molded part that can be manufactured from a plastic composition in accordance with the invention, in particular in the form of injection molded parts, foils, films, lacquers, coatings, foams, fibers, cables, tubes, sections, hollow bodies, ribbons, membranes, e.g. geo-membranes, lubricants, colorants, and/or adhesives that are manufactured by extrusion, injection molding, blow molding, calendering, pressing processes, spinning processes, and or rotomolding, e.g. for the electrical industry, for the construction industry, for the transport industry (automotive, aircraft, railroad), for medical applications, for domestic and electric appliances, for vehicle parts, for consumer products, for packaging, for furniture, and/or for textiles. The present invention further also relates to a stabilizer composition for the stabilization of thermoplastic plastic recyclates, preferably thermoplastic plastic recyclates, against oxidative, thermal, and/or actinic degradation, comprising (A) at least one substituted sugar, and (B) at least one primary antioxidant and/or at least one secondary antioxidant.

(124) All the preferred variants, exemplary embodiments, and comments (e.g. with respect to possible components and additives to be used) already described with respect to the method in accordance with the invention and with respect to the plastic composition in accordance with the invention also apply accordingly to the stabilizer composition in accordance with the invention.

(125) The present invention also relates to the use of at least one substituted sugar for the stabilization of thermoplastic plastic recyclates against oxidative, thermal, and/or actinic degradation.

(126) The present invention further relates to the use of the stabilizer composition in accordance with the invention or of a stabilizer composition comprising or consisting of (A) at least one substituted sugar and (B) at least one primary antioxidant and/or at least one secondary antioxidant for the stabilization of thermoplastics, preferably thermoplastic plastic recyclates, against oxidative, thermal, and/or actinic degradation.

(127) Preferred thermoplastics are here the same as those named with respect to the method in accordance with the invention and to the plastic composition in accordance with the invention.

(128) The stabilizer composition is particularly preferably used for the stabilization of polyolefin recyclates and against oxidative, thermal, and/or actinic degradation.

(129) All the preferred variants, exemplary embodiments, and comments (e.g. with respect to possible components and additives to be used) already described with respect to the method in accordance with the invention and with respect to the plastic composition in accordance with the invention and with respect to the stabilizer composition in accordance with the invention also apply

accordingly to the use of the stabilizer composition in accordance with the invention.

(130) The present invention will be described in more detail with reference to the following embodiments without restricting the invention to the specifically shown examples.

(131) A post-consumer polypropylene recycle from accumulator box ground stock (supplier: BSB Braubach) together with the additives given in the Table was conveyed in a circle in the melt at 210° C. in a twin screw microextruder (MC 5, manufacturer DSM) in continuous mode at 90 revolutions a minute for 30 minutes to check the effect of the stabilizer composition in accordance with the invention. The force absorption is measured in each case after 10, 20, and 30 minutes. The force is a measure for the toughness of the melt and thus for the molecular weight. The higher the remaining force (after a melting time of 2 minutes=100%), the smaller the degradation of the polymer.

(132) The addition of 0.1 to 0.3% of a stearyl substituted mannitol (>90% monosubstituted) results, in comparison with a recycle without additive, in increased processing stabilization, i.e. in a higher residual force. A further improvement of the processing stability is achieved in that 0.1% primary antioxidant (penta-erythritol-tetrakis[3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate] and 0.1% secondary antioxidant (tris-(2,4-di-tert-butylphenyl)phosphite) are added to 0.2% of the stearyl substituted mannitol.

(133) In an analog manner, the addition of the distearyl ester of the mucic acid in a concentration of 0.1 to 0.5%, in comparison with a polypropylene recycle without additive, results in increased processing stabilization, i.e. in a higher residual force after 30 minutes.

(134) Combinations of a stearyl substituted mannitol were further processed in a concentration of 0.2% in 1) LLDPE/LDPE film agglomerates from industrial collections together with 0.2% octadecyl-3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate at 210° C. 2) polyethylene/polypropylene mixture of treated recycle collections together with 0.1% pentaerythritol-tetrakis[3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate] and 0.1% (tris-(2,4-di-tert-butylphenyl)phosphite at 210° C. 3) polylactide (PLA) from used beverage cups together with 0.1% pentaerythritol-tetrakis[3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate] at 200° C. 4) polystyrene from recycle collections with 0.1% Wingstay L (reaction product) of 4-methylphenol with dicylopentadiene and isobutene) at 230° C.

Improved processing stabilization results in all cases in comparison with non-post stabilized recycle.

Claims

1. A method for stabilization of a thermoplastic plastic recycle against oxidative, thermal, and/or actinic degradation, comprising introducing component (A) into the thermoplastic plastic recycle, wherein the component (A) is at least one substituted sugar selected from the group consisting of an ester of an alditol having at least four carbon atoms, an ether of an alditol having at least four carbon atoms, an ester of a cyclitol, an ether of a cyclitol, an ester of a saccharic acid, an amide of an amino sugar, and a substituted amine of an amino sugar.

2. The method in accordance with claim 1, wherein the at least one substituted sugar comprises at least three free OH groups.

3. The method in accordance with claim 1, wherein the at least one substituted sugar is mono- or di-substituted.

4. The method in accordance with claim 1, wherein the alditol from which the ester or ether is formed has the molecular formula

$\text{HOCH}_2[\text{CH}(\text{OH})]_n\text{CH}_2\text{OH}$,

$\text{R}_1-\text{OCH}_2[\text{CH}(\text{OH})]_n\text{CH}_2\text{OH}$, or

$\text{HOCH}_2[\text{CH}(\text{OH})]_n[\text{CH}(\text{OR}_1)]\text{CH}_2\text{OH}$ where $n=2-6$, where R_1 is an optionally substituted sugar residue.

5. The method in accordance with claim 1, wherein the alditol from which the ester or ether is formed is selected from the group consisting of threitol, erythritol, galactinol, mannitol, ribitol, sorbitol, xylitol, arabitol, ismaltol, lactitol, maltitol, maltotritol, hydrated oligosaccharides, and polysaccharides with polyol end groups.
6. The method in accordance with claim 1, wherein the cyclitol from which the ester or ether is formed is selected from the group consisting of inositol, quercitol, viscumitol, bornesitol, conduritol, ononitol, pinitol, pinpollitol, ciceritol, quebrachitol, quinic acid, shikimic acid, and valienol.
7. The method in accordance with claim 1, wherein the saccharic acid from which the ester or ether is formed is selected from the group consisting of aldonic acids and aldaric acids.
8. The method in accordance with claim 1, wherein the saccharic amine from which the ester or ether is formed is selected from the group consisting of 1-amino-1-deoxyalditols and alpha-omega-diamino-alpha-omega-dideoxyalditols.
9. The method in accordance with claim 1, wherein the at least one substituted sugar is selected from the group consisting of the following compounds: $\text{---STR00031---STR00032---}$ where n is the same or different on every occurrence and is an integer between 1 and 100.
10. The method in accordance with claim 1, wherein 0.01 to 5.0 parts by weight of the at least one substituted sugar (component (A)) is introduced into 95.0 to 99.99 parts of the thermoplastic plastic recycle.
11. The method in accordance with claim 1, wherein, in addition to the at least one substituted sugar, at least one primary antioxidant and/or at least one secondary antioxidant (component (B)) is/are introduced into the thermoplastic plastic recycle.
12. The method in accordance with claim 11, wherein the at least one primary antioxidant is selected from the group consisting of phenolic antioxidants, amines, and lactones.
13. The method in accordance with claim 11, wherein the at least one secondary antioxidant is selected from the group consisting of phosphorus compounds and organo-sulfur compounds.
14. The method in accordance with claim 11, wherein at least one primary antioxidant and at least one secondary antioxidant are introduced into the thermoplastic plastic recycle.
15. The method in accordance with claim 11, wherein component (A) and component (B) are introduced into the thermoplastic recycle in a weight ratio of 95:5 to 5:95.
16. The method in accordance with claim 11, wherein component (A) and/or component (B) are introduced by mixing the component (A) and/or component (B) present as solids with the thermoplastic plastic recycle present as a solid, melting the mixture thereby produced, and subsequent cooling; or melting the component (A) and component (B) present as solids and introducing the melt thus produced into a melt of the thermoplastic plastic recycle.
17. The method in accordance with claim 11, wherein, with respect to 91 to 99.94 parts by weight of the thermoplastic plastic recycle, (A) 0.02 to 3 parts by weight of at least one substituted sugar (component (A)); and (b) 0.04 to 6 parts by weight of the at least one primary antioxidant and/or of at least one secondary antioxidant (component (B)) are introduced.
18. The method in accordance with claim 1, wherein the thermoplastic plastic recycle contains one or more of a) to n), which are as follows: a) recycled polymers of olefins and/or diolefins, b) recycled polystyrene, polymethyl styrene, polyvinyl naphthalene, styrene butadiene (SB), styrene butadiene styrene (SBS), styrene ethylene butylene styrene (SEBS), styrene ethylene propylene styrene, styrene isoprene, styrene isoprene styrene (SIS), styrene butadiene acrylonitrile (ABS), styrene acrylonitrile acrylate (ASA), styrene ethylene, styrene maleic acid anhydride polymers, graft copolymers of styrene on butadiene, maleic acid anhydride on SBS or SEBS, and graft copolymers of methyl methacrylate, styrene butadiene, and/or ABS, c) recycled polymers of unsaturated esters, d) recycled polymers of unsaturated alcohols and/or derivatives, e) recycled polyacetates and/or corresponding copolymers, f) recycled polyphenylene oxides and/or blends with polystyrene or polyamides, g) recycled polymers of cyclic ethers, h) recycled polyurethanes of

hydroxy terminated polyethers or polyesters and aromatic or aliphatic isocyanates, i) recycled polyamides, j) recycled polyamide imides, polyether imides, polyester imides, poly(ether)ketones, polysulfones, polyether sulfones, polyaryl sulfones, polyphenylene sulfides, polybenzimidazole azoles, and/or polyhydantoines, k) recycled polyesters of aliphatic or aromatic dicarboxylic acids and diols or of hydroxy carboxylic acids, l) recycled polycarbonates, polyester carbonates, and/or blends thereof, m) recycled halogenated polymers, and n) recycled cellulose derivatives.

19. The method in accordance with claim 18, wherein the recycled polymers of olefins and diolefins are selected from the group consisting of polyethylene, polypropylene, polyisobutylene, poly-4-methyl-pentene-1, polybutadiene, polyisoprene, polycyclooctene, polyalkylene-carbon monoxide copolymers, EPM, EPDM, ethylene-vinyl acetate (EVA), ethylene-acrylic esters, ethylene-acrylic acid glycidyl acrylate, polypropylene-g-maleic acid anhydride, polypropylene-g-acrylic acid, and polyethylene-g-acrylic acid.

20. The method in accordance with claim 19, wherein the polyethylene is selected from the group consisting of LDPE, LLDPE, VLDPE, ULDPE, MDPE, HDPE, UHMWPE, and metallocene PE (m-PE).

21. A plastic composition comprising at least one substituted sugar and at least one thermoplastic recyclate, wherein the at least one substituted sugar is selected from the group consisting of an ester of an alditol having at least four carbon atoms, an ether of an alditol having at least four carbon atoms, an ester of a cyclitol, an ether of a cyclitol, an ester of a saccharic acid, an amide of an amino sugar, and a substituted amine of an amino sugar.

22. The plastic composition in accordance with claim 21, wherein the plastic composition comprises 0.01 to 5.0 parts by weight of the at least one substituted sugar and 95.0 to 99.99 parts by weight of the at least one thermoplastic plastic recyclate.

23. The plastic composition in accordance with claim 21, further comprising at least one primary antioxidant and/or at least one secondary antioxidant.

24. The plastic composition in accordance with claim 23, wherein the plastic composition comprises 0.02 to 3 parts by weight of the at least one substituted sugar, 0.04 to 6 parts by weight of the at least one primary antioxidant and/or the at least one secondary antioxidant, and 91 to 99.94 parts by weight of the at least one thermoplastic plastic recyclate.

25. The plastic composition in accordance with claim 23, wherein the plastic composition comprises 0.02 to 3 parts by weight of the at least one substituted sugar, 0.02 to 3 parts by weight of the at least one primary antioxidant, 0.02 to 3 parts by weight of the at least one secondary antioxidant, and 91 to 99.94 parts by weight of the at least one thermoplastic plastic recyclate.

26. The plastic composition in accordance with claim 21, wherein the plastic composition additionally comprises at least one additive selected from the group consisting of UV absorbers, light stabilizers, metal deactivators, filler deactivators, antiozonants, nucleation agents, anti-nucleation agents, impact strength enhancers, plasticizers, lubricants, rheology modifiers, thixotropic agents, chain extenders, optical brighteners, antimicrobial active agents, antistatic agents, slip agents, anti-blocking agents, coupling agents, crosslinking agents, anti-cross-linking agents, hydrophilization agents, hydrophobing agents, bonding agents, dispersing agents, compatibilizers, oxygen scavengers, acid scavengers, expanding agents, degradation additives, defoaming agents, odor scavengers, marking agents, anti-fogging agents, fillers, and reinforcement agents.

27. The plastic composition in accordance with claim 21, wherein the plastic composition additionally comprises at least one additive that is selected from the group consisting of a) acid scavengers, b) light stabilizers, c) dispersants, and d) filler deactivators.

28. A molded part comprising the plastic composition of claim 21, wherein the molded part is in a form selected from the group consisting of an injection molded part, a foil, a film, a lacquer, a coating, a foam, a fiber, a cable, a tube, a section, a hollow body, a ribbon, and a membrane,

wherein the molded part is manufactured by a process selected from the group consisting of extrusion, injection molding, blow molding, calendering, pressing, spinning, and rotomolding.
